

```

Lab 5 a;
Var 11;
1 a;
a = -1; b = 1;
f[x_] = Cos[x]^3;
n = 5;

Do[{x_i = Cos[Pi*(2*i+1)/(2*(n+1))], f_i = f[x_i]}, {i, 0, n}]

P[x_] = Sum[Product[Pi*(2*i+1)/(2*(n+1)), {i, 0, k-1}], {k, 0, n}] * f_k // Simplify // Expand

-1/3 Cos[1/sqrt(2)]^3 + 16/3 x^2 Cos[1/sqrt(2)]^3 - 16/3 x^4 Cos[1/sqrt(2)]^3 + 2/3 Cos[(-1+sqrt(3))/(2*sqrt(2))]^3 + Cos[(-1+sqrt(3))/(2*sqrt(2))]^3/sqrt(3) -
8/3 x^2 Cos[(-1+sqrt(3))/(2*sqrt(2))]^3 - 2 x^2 Cos[(-1+sqrt(3))/(2*sqrt(2))]^3/sqrt(3) + 8/3 x^4 Cos[(-1+sqrt(3))/(2*sqrt(2))]^3 + 2/3 Cos[(1+sqrt(3))/(2*sqrt(2))]^3 -
Cos[(1+sqrt(3))/(2*sqrt(2))]^3/sqrt(3) - 8/3 x^2 Cos[(1+sqrt(3))/(2*sqrt(2))]^3 + 2 x^2 Cos[(1+sqrt(3))/(2*sqrt(2))]^3/sqrt(3) + 8/3 x^4 Cos[(1+sqrt(3))/(2*sqrt(2))]^3

Tbl = Table[{x_i, f_i}, {i, 0, n}];
Pl[x_] = InterpolatingPolynomial[Tbl, x] // Simplify // Expand

-1/3 Cos[1/sqrt(2)]^3 + 16/3 x^2 Cos[1/sqrt(2)]^3 - 16/3 x^4 Cos[1/sqrt(2)]^3 + 2/3 Cos[(-1+sqrt(3))/(2*sqrt(2))]^3 + Cos[(-1+sqrt(3))/(2*sqrt(2))]^3/sqrt(3) -
8/3 x^2 Cos[(-1+sqrt(3))/(2*sqrt(2))]^3 - 2 x^2 Cos[(-1+sqrt(3))/(2*sqrt(2))]^3/sqrt(3) + 8/3 x^4 Cos[(-1+sqrt(3))/(2*sqrt(2))]^3 + 2/3 Cos[(1+sqrt(3))/(2*sqrt(2))]^3 -
Cos[(1+sqrt(3))/(2*sqrt(2))]^3/sqrt(3) - 8/3 x^2 Cos[(1+sqrt(3))/(2*sqrt(2))]^3 + 2 x^2 Cos[(1+sqrt(3))/(2*sqrt(2))]^3/sqrt(3) + 8/3 x^4 Cos[(1+sqrt(3))/(2*sqrt(2))]^3

P[x] == Pl[x]

True

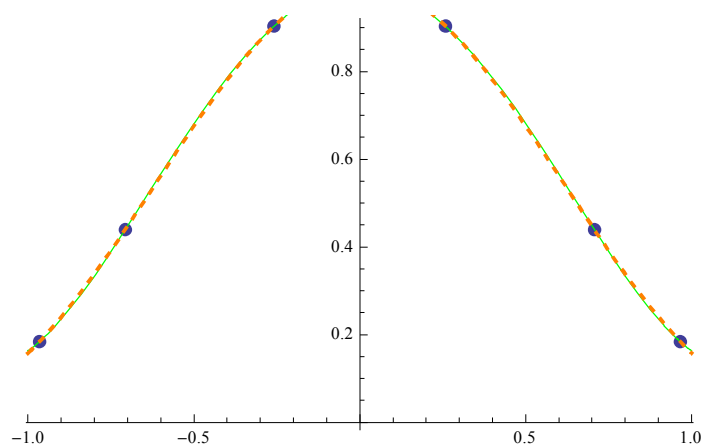
2 a;
M1 = Maximize[{D[f[x], {x, n+1}], a <= x <= b}, x][[1]];
M2 = Minimize[{D[f[x], {x, n+1}], a <= x <= b}, x][[1]];
M = Max[Abs[M1], Abs[M2]];
R = M/(n+1)! 2^n // N
0.00794271

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3 a;
Gr1 = ListPlot[Tbl, PlotStyle -> {PointSize[0.02]}];
Gr2 = Plot[P[x], {x, a, b}, PlotStyle -> {Green}];
Gr3 = Plot[f[x], {x, a, b}, PlotStyle -> {Orange, Dashed, Thick}];
Show[Gr1, Gr2, Gr3]

```



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2 a;
f[x_] = Cos[2.2 x];
a = 0; b = 1; eps = 10-5;

M1 := Maximize[{D[f[x], {x, n + 1}], a ≤ x ≤ b}, x][[1]]
M2 := Minimize[{D[f[x], {x, n + 1}], a ≤ x ≤ b}, x][[1]]
M := Max[Abs[M1], Abs[M2]]
n = 1;

While[ $\left(R = \frac{M (b - a)^{n+1}}{(n + 1)! 2^{2n+1}}\right) > \text{eps}, n++]$ 

n
6

R <= eps

True

Do[ $\left\{x_i = \frac{(b - a)}{2} * \text{Cos}\left[\frac{\text{Pi} * (2 * i + 1)}{2 * (n + 1)}\right] + \frac{a + b}{2}, f_i = f[x_i]\right\}, \{i, 0, n\}]$ 

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$$P[x_] = \sum_{k=0}^n \left(\prod_{i=0}^{k-1} \frac{x - x_i}{x_k - x_i} \right) * \left(\prod_{i=k+1}^n \frac{x - x_i}{x_k - x_i} \right) * f_k \text{ // Expand // N}$$

$$1. - 0.0004671 x - 2.41261 x^2 - 0.0432635 x^3 + 1.09414 x^4 - 0.157532 x^5 - 0.0687728 x^6$$

```

Tbl = Table[{xi, fi}, {i, 0, n}];

Gr1 = ListPlot[Tbl, PlotStyle -> {PointSize[0.02]}];
Gr2 = Plot[P[x], {x, a, b}, PlotStyle -> {Green}];
Gr3 = Plot[f[x], {x, a, b}, PlotStyle -> {Orange, Dashed, Thick}];
Show[Gr1, Gr2, Gr3]

```

